SWAROOP SRIDHAR

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EDUCATION

STEVENS INSTITUTE OF TECHNOLOGY — Hoboken, NJ
Master of Science in Computer Science — Expected Dec 2027

RENSSELAER POLYTECHNIC INSTITUTE — Troy, NY
Bachelor of Science in Computer Science; Bachelor of Science in Information Technology and Web Science — May 2026
Concentration: Machine Learning
Awards: 3rd Place, RPI Quantum Hackathon 2026; Best AI Hack, HackRPI 2025; Dean's List: Fall 2024, Spring 2026

CORE SKILLSETS

- **Programming Languages:** Python3, C++/C, Java, JavaScript, TypeScript, SQL, HTML, CSS, Assembly, Haskell, Erlang
- **ML/AI & Data Science:** PyTorch, Scikit-learn, NumPy, SciPy, pandas, Matplotlib, OpenCV, Optuna, NLTK; Machine Learning, Deep Learning, Computer Vision
- **Software & Tools:** Flask, React.js, Node.js, REST APIs, CustomTkinter, Full Stack Development, OOP, Data Structures, Algorithms, Git, GitHub, Docker, Linux, Microsoft Azure, CI/CD, Unit Testing, Agile/Scrum

WORK EXPERIENCE

SCIENTIFIC COMPUTATION RESEARCH CENTER — Troy, NY
Research Assistant — Jun 2026 – Present

- Rebuilt the classifier's evaluation from pixel- to point-level detection, improving held-out F1 from 0.61 to 0.72 and exposing that the old metric overstated accuracy.
- Replaced binary-mask targets with CenterNet-style Gaussian heatmaps and a focal loss to sharpen X-point localization.
- Built a modular CSV-based pipeline for peak extraction and point matching, enabling independent model re-scoring.

Undergraduate Research Assistant — Jun 2025 – May 2026

- Resolved a model convergence failure, cutting training loss from 0.94 to 0.34 via residual skip connections, batch normalization, and feature normalization.
- Set up a CI pipeline with unit tests covering loss functions, dataset validation, and model checkpointing.
- Accelerated training 1.4x with bfloat16 mixed precision, reducing training time 28% and GPU memory 22%.

MACHINE LEARNING INTERN — May 2026 – Aug 2026
Center For Disability Services – Incubation Center — Watervliet, NY

- Standardized 344K+ 2D sensor maps into a common format, enabling consistent analysis across multiple sources.
- Benchmarked five regressors under subject-grouped cross-validation, finding peak-pressure accuracy collapsed from 43% to 10% on unseen subjects, with mean error within 1% of a mean-position baseline.
- Ran 13 known-load trials on a research pressure mat, showing its error tracked contact area and surface compliance rather than a fixed offset, with real seated load under-reported 21%.
- Wrote reusable command-line tools that validate sensor-map exports and generate versioned JSON summaries, backed by regression tests for core analysis outputs.

APPLIED MACHINE LEARNING RESEARCHER — Jun 2025 – Apr 2026
Project Voice (CINPCT 26) - Rensselaer Polytechnic Institute — Troy, NY

- Built a GMM data-synthesis pipeline that expanded 100 physiological observations to 5,000 records with no significant KS-test differences across 29 numeric features.
- Benchmarked nine ML and deep-learning models, with LSTM achieving 70.8% accuracy on a 500-case synthetic test set.
- Compared five synthesis methods, finding correlation preservation predicted transfer accuracy better than marginal fidelity: GMM preserved the correlation matrix 21× better than CTGAN and transferred at 66% vs 32%.

PROJECTS

DELOITTE CAPSTONE – ENTERPRISE RAG SEARCH ENGINE — Feb 2026 – Apr 2026

- Built a RAG pipeline (Next.js, Python FastAPI, ChromaDB, Google Gemini) that returns LLM-generated answers with source citations across a multi-format document corpus.
- Designed the retrieval layer around ContextualCompressionRetriever and LCEL, with LLM-based query rewriting via Gemini to improve retrieval precision before each search.
- Demoed the system to the client team, running live queries that returned confidence-ranked source documents alongside a cited LLM answer, with role-based access control applied to results.

FYTÓSPOT – PLANT IDENTIFICATION SYSTEM — Jan 2025 – Mar 2025

- Built a PyTorch ResNet-50 pipeline for plant-species identification across 1,081 classes, surfacing per-prediction confidence scores in the desktop and web UIs.
- Developed real-time OpenCV capture and tracking in a CustomTkinter desktop app, with a Flask web UI for single-image detection and identification.